

Central Valley Regional Water Quality Control Board

20 July 2018

Jerry Marcotte
Senior Sanitary Engineer
California Department of Transportation
1801 30th Street, MS 9-3/11H
Sacramento, CA 95816-8041

CERTIFIED MAIL
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NOTICE OF APPLICABILITY (NOA), STATE WATER RESOURCES CONTROL BOARD ORDER WQ 2014-0153-DWQ, GENERAL WASTE DISCHARGE REQUIREMENTS FOR SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS, CALIFORNIA DEPARTMENT OF TRANSPORTATION DISTRICT 10, JOHN "CHUCK" ERRECA SAFETY ROADSIDE REST AREA, MERCED COUNTY

On 22 December 2017, the California Department of Transportation, District 10 (Caltrans or Discharger), submitted a Report of Waste Discharge (RWD) for a wastewater treatment system at its John "Chuck" Erreca Safety Roadside Rest Area (SRRA) facility (Facility) about 18 miles west of the City of Firebaugh along Interstate 5 in Merced County. Caltrans also submitted a 17 November 2017 Title 22 Engineering Report for a recycled water project at the facility, which was later approved by a 12 February 2018 Division of Drinking Water letter. Based on the information provided, the system treats and disposes of less than 100,000 gallons of wastewater per day, and is therefore eligible for coverage under the general and specific conditions of State Water Resources Control Board (State Water Board) Water Quality Order 2014-0153-DWQ *General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems* (General Order). This letter serves as formal notice that the General Order is applicable to your system and the wastewater discharge described below. You are hereby assigned General Order **2014-0153-DWQ-R5278** for your system.

You should familiarize yourself with the entire General Order and its attachments enclosed with this letter, which describe mandatory discharge and monitoring requirements. Sampling, monitoring, and reporting requirements applicable to your treatment and disposal methods must be completed in accordance with the appropriate treatment system sections of the General Order and the attached *Monitoring and Reporting Program* (MRP) No. 2014-0153-DWQ-R5278. This MRP was developed after consideration of your waste characterization and site conditions described in the attached memorandum.

DISCHARGE DESCRIPTION

Caltrans owns and operates the Erreca SRRA in Sections 20 and 29, T12S, R11E MDB&M in Merced County. The Facility currently treats wastewater in two separate systems on the

northbound and southbound sides of I-5 each consisting of eight 6,000 gallon septic tanks with leachfield disposal. Wastewater flow is approximately 8,280 gpd (90% of potable use).

The RWD proposes an automated wastewater treatment system to provide disinfected tertiary-treated wastewater for non-potable use (toilet flushing). The proposed system will recycle approximately 80-90% of the wastewater flow with 10-20% being dispersed to an existing leachfield on the northbound side of Interstate-5.

The proposed wastewater treatment system will consist of septic tanks, upflow anaerobic filters, backwashing filters, a textile media filter, a microfiber filter, ozone injection, and chlorine disinfection. Excess or off-spec wastewater can be diverted to the leachfield after the septic tanks, upflow filters, or after the chlorine contact basin.

The disposal area consists of four leachfield areas, each with approximately 1,000 linear feet of dispersal laterals. The leachfield trenches consist of 2-foot-wide trenches with perforated pipe set in a bed of sand at a minimum depth of 2 feet. The system is equipped with a dosing pump to alternate pumping between each of the leachfield areas.

FACILITY SPECIFIC REQUIREMENTS AND EFFLUENT LIMITATIONS

The Discharger will maintain exclusive control over the discharge and shall comply with the terms and conditions of this NOA, General Order 2014-0153-DWQ, with all attachments, and MPR No. 2014-0153-DWQ-R5278.

In accordance with Section B.1 of the General Order, treated wastewater discharged to the Facility's leachfield **shall not exceed 20,000 gpd as a monthly average**.

The General Order states in Section B.1.I that the Discharger shall comply with the setbacks described in Table 3. This table summarizes different setback requirements for wastewater system equipment, activities, land application areas, and storage and/or treatment ponds from sensitive receptors and property lines where applicable. The Discharger shall comply with the applicable setback requirements, as summarized in the following table:

Site Specific Applicable Setback Requirements			
Equipment or Activity	Domestic Well	Property Line	Flowing Stream
Septic Tank, Treatment System, Collection System ^a	150 ft. ^b	5 ft. ^c	50 ft. ^c
Leachfield ^d	100 ft. ^{c,e}	5 ft. ^c	100 ft. ^c

a. Septic Tank, Treatment System, or Collection System addresses equipment located below ground or that impedes leak detection by routine visual inspection.

b. Setback established by Onsite Wastewater Treatment System Policy, section 7.5.6.

c. Setback established by California Plumbing Code, Table K-1.

d. Leachfield includes all subsurface dispersal systems, including mound systems except seepage pits.

e. California Well Standards, part II, section 8. Site-specific conditions may allow reduced setback or require an increased setback. See discussion in Well Standards.

The Discharger shall comply with the septic system requirement specified in Section B.2 of the General Order. The General Order states in Section B.2.c that to the maximum extent possible,

RV, portable toilet, or similar wastes shall not be discharged to a septic tank or functionally equivalent system (e.g., Imhoff tank) without subsequent additional treatment (e.g., aerated pond, recirculating sand filter, etc.) prior to disposal.

The General Order states in Section B.2.d that septic tanks shall be pumped when any of the following conditions exists:

- i. The combined thickness of sludge and scum exceeds one-third of the tank depth of the first compartment.
- ii. The scum layer is within 3 inches of the outlet device.
- iii. The sludge layer is within 8 inches of the outlet device.

The General Order includes subsurface disposal system requirements in Section B.6. The Facility includes leachfields; therefore, the Discharger must comply with the requirements in Section B.6. Based on the information provided by the Discharger, the Facility's leachfields are classified as Class V wells. Therefore, the Discharger must comply with USEPA Underground Injection Control requirements as specified in Section B.6.g. of the General Order.

The General Order includes recycled water system requirements in Section B.7. In addition to compliance with the requirements of Section B.7, the Discharger shall comply with the permit conditions specified in the 12 February 2018 Division of Drinking Water Title 22 Engineering Report approval letter, these include:

- The Discharger shall reclaim the treated effluent in the manner described in the 17 November 2017 Title 22 Engineering Report. Any changes must be approved by the Division of Drinking Water,
- Compliance with the 8 June 2008 letter "Use of Amiad Filtration Systems AMF Wastewater Filter to comply with the California Water Recycling Criteria",
- Submit quarterly reports to DDW with the required information (see MRP), and
- Provide proper notification when certain conditions occur at the Facility (see attached 12 February 2018 Division of Drinking Water Letter)

Failure to comply with the requirements in this NOA, General Order 2014-0153-DWQ, with all attachments, and MRP No. 2014-0153-DWQ-R5278 could result in an enforcement action as authorized by provisions of the California Water Code. Discharge of wastes other than those described in this NOA is prohibited. If the method of waste disposal changes from that described in this NOA, you must submit a new Report of Waste Discharge describing the new operation. If flow to the Facility substantially increases and approaches 20,000 gpd, you must contact Central Valley Water Board staff to determine if further analysis is required.

Provision E.1 of the General Order requires dischargers enrolled under the General Order to prepare and implement the following reports within **90 days** of the issuance of this NOA (**By 19 October 2018**):

- Spill Prevention and Emergency Response Plan (Provision E.1.a)
- Sampling Analysis Plan (Provision E.1.b)
- Sludge Management Plan (Provision E.1.c)

The General Order requires the Sludge Management Plan to be submitted to the Central Valley Water Board within 90 days of the issuance of this NOA.

As stated in Section E.2.w., in the event any change in control or ownership of the facility or wastewater disposal areas, the Discharger must notify the succeeding owner or operator of the existence of this General Order by letter, a copy of which shall be immediately forwarded to the Central Valley Water Board Executive Officer.

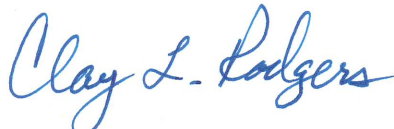
The required annual fee specified in the annual billing from the State Water Board shall be paid until this NOA is officially terminated. You must notify this office in writing if the discharge regulated by the General Order ceases, so that we may terminate coverage and avoid unnecessary billing.

The Central Valley Water Board has gone to a Paperless Office System. All regulatory documents, submissions, materials, data, monitoring reports, and correspondence should be converted to a searchable Portable Document Format (PDF) and submitted electronically. Documents that are less than 50 MB should be emailed to: centralvalleyfresno@waterboards.ca.gov. Documents that are 50 MB or larger should be transferred to a disk and mailed to the Central Valley Water Board office at 1685 E Street, Fresno, CA 93706. To ensure that your submittals are routed to the appropriate staff, the following information block should be included in any email used to transmit documents to this office: Program: Non-15, WDID: 5B24NC00381, Facility Name: John "Chuck" Erreca SRRA, Order: 2014-0153-DWQ-R5278.

In order to conserve paper and reduce mailing costs, a paper copy of the General Order has been sent only to the Discharger. Others are advised that the General Order is available on the State Water Board's web site at:

http://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2014/wqo2014_0153_dwq.pdf

If you have any questions regarding this matter, please contact Daniel Benas by phone at (559) 445-5500 or email at Daniel.Benas@waterboards.ca.gov.



for Patrick Pulupa
Executive Officer

Attachments: Attachment A – Facility Map
Attachment B – Process Flow Diagram
State Water Resources Control Board Order WQ 2014-0153-DWQ
(Discharger Only)
Monitoring and Reporting Program No. 2014-0153-DWQ-R5278
Review Memorandum of California Department of Transportation Report of
Waste Discharge
12 February 2018 Division of Drinking Water Approval Letter

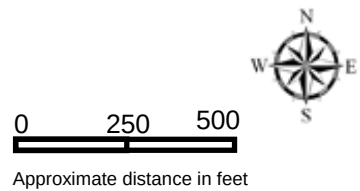
cc: Merced County Environmental Health, Merced
Kassy Chauhan, Division of Drinking Water, Fresno

ATTACHMENT A – FACILITY MAP
CALIFORNIA DEPARTMENT OF TRANSPORTATION
JOHN “CHUCK” ERRECA SRRA

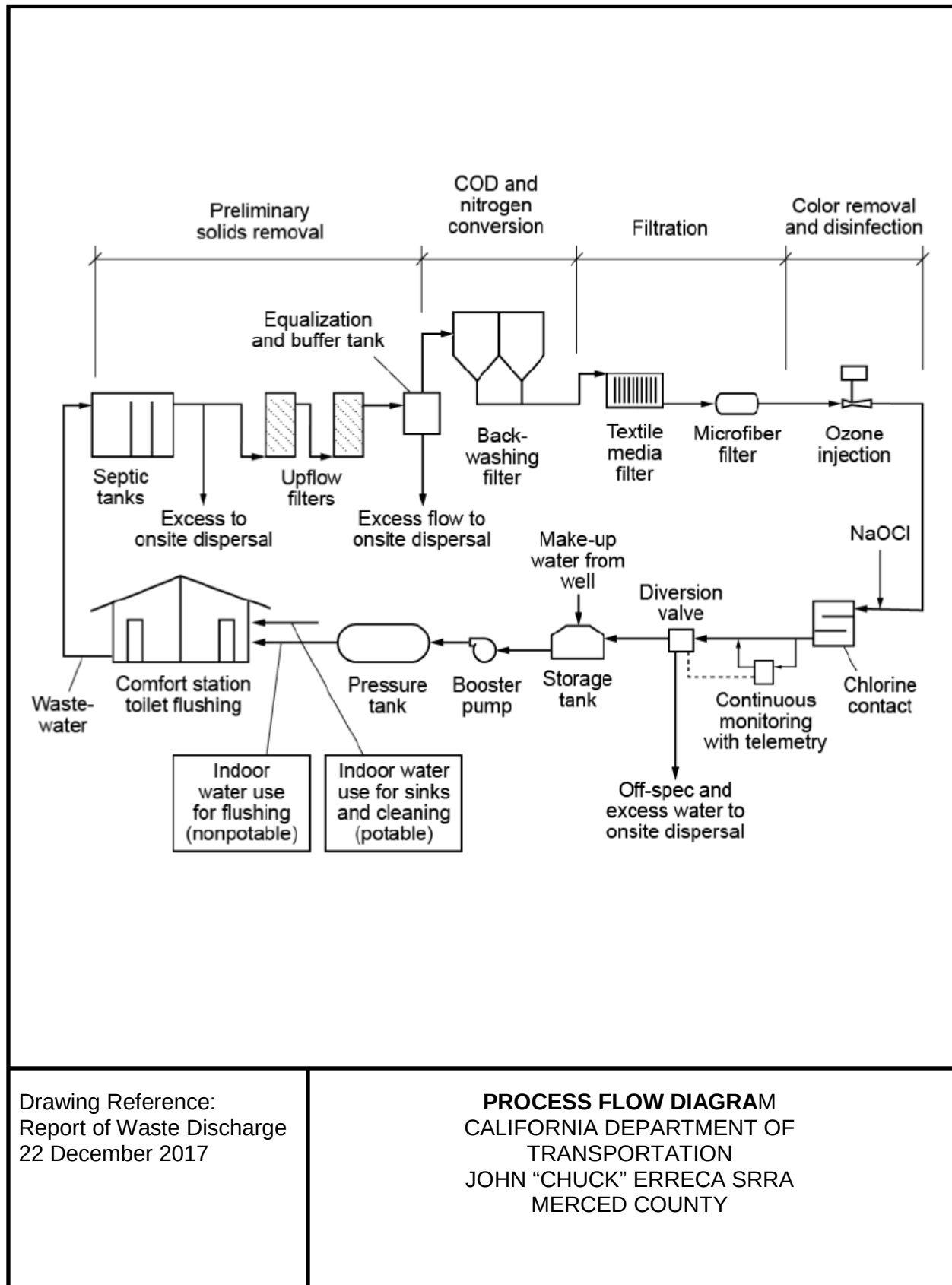


Drawing Reference:
Google Earth
Map Data: © 2018

Attachment A – Facility Map
California Department of
Transportation
John “Chuck” Erreca SRRA
Merced County



ATTACHMENT B – PROCESS FLOW DIAGRAM
 CALIFORNIA DEPARTMENT OF TRANSPORTATION
 JOHN “CHUCK” ERRECA SRRA



Drawing Reference:
 Report of Waste Discharge
 22 December 2017

PROCESS FLOW DIAGRAM
 CALIFORNIA DEPARTMENT OF
 TRANSPORTATION
 JOHN “CHUCK” ERRECA SRRA
 MERCED COUNTY

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 2014-0153-DWQ-R5278

FOR

CALIFORNIA DEPARTMENT OF TRANSPORTATION DISTRICT 10
JOHN "CHUCK" ERRECA SRRA
ONSITE WASTEWATER TREATMENT SYSTEM
MERCED COUNTY

This Monitoring and Reporting Program (MRP) describes requirements for monitoring a wastewater treatment system. This MRP is issued pursuant to Water Code section 13267. The California Department of Transportation, District 10 (Discharger) shall not implement any changes to this MRP unless and until a revised MRP is issued by the Regional Water Quality Control Board, Central Valley Region (Central Valley Water Board) or the Executive Officer.

Water Code section 13267 states, in part:

"In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports."

Water Code section 13268 states, in part:

"(a)(1) Any person failing or refusing to furnish technical or monitoring program reports as required by subdivision (b) of section 13267, or failing or refusing to furnish a statement of compliance as required by subdivision (b) of section 13399.2, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in accordance with subdivision (b).

(b)(1) Civil liability may be administratively imposed by a regional board in accordance with article 2.5 (commencing with section 13323) of chapter 5 for a violation of subdivision (a) in an amount which shall not exceed one thousand dollars (\$1,000) for each day in which the violation occurs."

The Discharger owns and operates the John "Chuck" Erreca Safety Roadside Rest Area (SRRA) wastewater system (Facility) that is subject to the Notice of Applicability (NOA) of Water Quality Order 2014-0153-DWQ-R5278. The reports are necessary to ensure that the Discharger complies with the NOA and General Order. Pursuant to Water Code section 13267, the Discharger shall implement this MRP and shall submit the monitoring reports described herein.

All samples shall be representative of the volume and nature of the discharge or matrix of material sampled. The name of the sampler, sample type (grab or composite), time, date, location, bottle type, and any preservative used for each sample shall be recorded on the sample chain of custody form. The chain of custody form must also contain all custody information including date, time, and to whom samples were relinquished. If composite samples are collected, the basis for sampling (time or flow weighted) shall be approved by Central Valley Water Board staff.

Field test instruments (such as those used to test pH, dissolved oxygen, and electrical conductivity) may be used provided that they are used by a State Water Resources Control Board, Environmental Laboratory Accreditation Program certified laboratory, or:

1. The user is trained in proper use and maintenance of the instruments;
2. The instruments are field calibrated prior to monitoring events at the frequency recommended by the manufacturer;
3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
4. Field calibration reports are maintained and available for at least three years.

If monitoring consistently shows no significant variation in magnitude of a constituent concentration or parameter after at least 12 months of monitoring, the Discharger may request this MRP be revised to reduce monitoring frequency. The proposal must include adequate technical justification for reduction in monitoring frequency.

SEPTIC TANK MONITORING

Monitoring of septic tanks shall include the following:

<u>Parameter</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow Rate	Gallons per day	Metered ¹	Continuous	Annually

¹. Flow rate may be metered or estimated based on potable water supply meter readings or other approved method.

Septic tanks shall be inspected and/or pumped at least as frequently as described below. Inspections of sludge and scum depth are not required if the tanks are pumped at least annually.

<u>Parameter</u>	<u>Units</u>	<u>Measurement Type</u>	<u>Inspection/Reporting Frequency</u>
Sludge depth and scum thickness in each compartment of each tank	Feet	Staff Gauge	Annually
Distance between bottom of scum layer and bottom of outlet device	Inches	Staff Gauge	Annually
Distance between top of sludge layer and bottom of outlet device	Inches	Staff Gauge	Annually
Effluent filter condition (if equipped, clean as needed)	NA ¹	NA ¹	Annually
Electrical Conductivity	µmhos/cm	Grab ²	1/Month
Total Nitrogen	mg/L	Grab ²	1/Quarter

1. NA = not applicable

2. Grab samples for electrical conductivity and total nitrogen shall be taken at a location prior to discharge to the leachfield.

Septic tanks shall be pumped when any one of the following conditions exists:

1. The combined thickness of sludge and scum exceeds one-third of the tank depth of the first compartment.
2. The scum layer is within 3 inches of the outlet device.
3. The sludge layer is within 8 inches of the outlet device.

If a septic tank is pumped during the year, the pumping report shall be submitted with the annual report. All pumping reports shall be submitted with the next regularly scheduled monitoring report. At a minimum, the record shall include the date, nature of service, service company name, and service company license number.

SUBSURFACE DISPOSAL AREA

Monitoring shall be sufficient to determine if wastewater is evenly applied, the disposal area is not saturated, burrowing animals and/or deep rooted plants are not present, and odors are not present. Inspection of dosing pump controllers, automatic distribution valves, etc. is required to maintain optimum treatment in the disposal area. Monitoring of the Facility's leachfield system shall include, at a minimum, the following:

<u>Constituent</u>	<u>Inspection Frequency</u>	<u>Reporting Frequency</u>
Pump Controllers, Automatic Valves, etc. ¹	Quarterly	Quarterly
Nuisance Odor Condition	Quarterly	Quarterly
Saturated Soil Conditions ²	Quarterly	Quarterly
Plant Growth ³	Quarterly	Quarterly
Vectors or Animal Burrowing ⁴	Quarterly	Quarterly

1. All pump controllers and automatic distribution valves shall be inspected for proper operation as recommended by the manufacturer.
2. Inspect a disposal area for saturated conditions.
3. Shallow-rooted plants are generally desirable, deep-rooted plants such as trees shall be removed as necessary.
4. Evidence of animals burrowing shall be immediately investigated and burrowing animal populations controlled as necessary.

RECYCLED WATER MONITORING

Monitoring of the recycled water system shall be after the chlorine disinfection system and shall include:

Constituent/Parameter	Units	Sample Frequency	Reporting Frequency
Total Coliform ¹	MPN/100 mL	Daily	Quarterly
Chlorine residual	mg/L	Continuous	Quarterly
Turbidity	NTU	Continuous	Quarterly

¹. The Discharger shall conduct total coliform monitoring daily when recycling treated effluent at the Facility.

SOLIDS DISPOSAL MONITORING

The Discharger shall report the handling and disposal of all solids (e.g., screenings, grit, sludge, biosolids, etc.) generated at the wastewater system. Records shall include the name/contact information for the hauling company, the type and amount of waste transported, the date removed from the wastewater system, the disposal facility name and address, and copies of analytical data required by the entity accepting the waste. These records shall be submitted as part of the annual monitoring report.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, solids, etc.), and reported analytical or visual inspection results are readily discernible. The data shall be summarized to clearly illustrate compliance with the General Order and NOA as applicable. The results of any monitoring done more frequently than required at the locations specified in the MRP shall be reported in the next regularly scheduled monitoring report and shall be included in calculations as appropriate.

The Central Valley Water Board has gone to a Paperless Office System. All regulatory documents, submissions, materials, data, monitoring reports, and correspondence should be converted to a searchable Portable Document Format (PDF) and submitted electronically. Documents that are less than 50MB should be emailed to: centralvalleyfresno@waterboards.ca.gov. Documents that are 50MB or larger should be transferred to a disk and mailed to the appropriate Regional Water Board office, in this case 1685 E Street, Fresno, CA 93706. To ensure that your submittals are routed to the appropriate staff, the following information block should be included in any email used to transmit documents to this office: Program: Non-15, WDID: 5B24NC00381, Facility Name: John "Chuck" Erreca SRRA, Order: 2014-0153-DWQ-R5278.

A. Quarterly Monitoring Reports

Quarterly reports shall be submitted to the Central Valley Water Board and the State Water Resources Control Board, Division of Drinking Water on the **first day of the second month after the quarter ends** (e.g., the January-March Quarterly Report is due by May 1st). The reports shall bear the certification and signature of the Discharger's authorized representative. At a minimum, the quarterly reports shall include:

1. Results of all required monitoring.
2. A comparison of monitoring data to the discharge specifications, disclosure of any violations of the NOA and/or General Order, and an explanation of any violation of those requirements. (Data shall be presented in tabular format.)
3. Report the running 7-day median calculation and maximum daily total coliform reading for each month.
4. Report the minimum daily chlorine residual and minimum daily chlorine contact time (CT).
5. Report average filter effluent turbidity (24 hour period), 95th percentile filter effluent turbidity (24 hour period), and daily maximum turbidity reading when the plant is operating.
6. If requested by staff, copies of laboratory analytical report(s) and chain of custody form(s).

B. Annual Report

Annual Reports shall be submitted to the Central Valley Water Board by **March 1st following the monitoring year**. The Annual Report shall include the following:

1. Tabular and graphical summaries of all monitoring data collected during the year.
2. An evaluation of the performance of the wastewater treatment system, including discussion of capacity issues, nuisance conditions, system problems, and a forecast of the flows anticipated in the next year. A flow rate evaluation, as described in the General Order (Provision E.2.c), shall also be submitted.
3. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into compliance with the NOA and/or General Order.
4. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.
5. The name and contact information for the wastewater operator responsible for operation, maintenance, and system monitoring.

A letter transmitting the monitoring reports shall accompany each report. The letter shall report violations found during the reporting period, and actions taken or planned to correct the violations and prevent future violations. The transmittal letter shall contain the following penalty of perjury statement and shall be signed by the Discharger or the Discharger's authorized agent:

"I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of the those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

The Discharger shall implement the above monitoring program as of the date of this MRP.

Ordered by:


for PATRICK PULUPA, Executive Officer

DATE

Central Valley Regional Water Quality Control Board

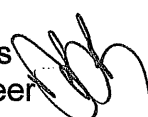
TO: Scott J. Hatton
Supervising Engineer
RCE 67889



FROM: Alexander S. Mushegan
Senior Engineer
RCE 84208



Daniel Benas
WRC Engineer



DATE: 20 July 2018

**SUBJECT: APPLICABILITY OF COVERAGE UNDER STATE WATER
RESOURCES CONTROL BOARD ORDER WQ 2014-0153-DWQ,
GENERAL WASTE DISCHARGE REQUIREMENTS FOR SMALL
DOMESTIC WASTEWATER TREATMENT SYSTEMS, CALIFORNIA
DEPARTMENT OF TRANSPORTATION DISTRICT 10, JOHN
"CHUCK" ERRECA SAFETY ROADSIDE REST AREA, MERCED
COUNTY**

On 22 December 2017, Central Valley Water Board staff (staff) received a Report of Waste Discharge (RWD) for a wastewater treatment system at the John "Chuck" Erreca Safety Roadside Rest Area (SRRA). The SRRA is located on both the north and southbound sides of Interstate 5 in Merced County, about 18 miles south of Los Banos (Sections 20 and 29, Township 12 South, Range 11 East MDB&M). The RWD includes a Form 200; applicable filing fee; technical report prepared by Manjunath Akkipeddi, P.E. and Jerry Marcotte, P.E. with Caltrans; and the Categorical Exemption/Categorical Exclusion Determination CEQA document. This memorandum provides a summary of staff's review of the RWD and the applicability of this discharge to be covered under State Water Resources Control Board Order WQ 2014-0153-DWQ, General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems (General Order).

DESCRIPTION OF DISCHARGE

The existing onsite wastewater treatment system was permitted by Merced County and consists of two separate systems, one on the northbound side of I-5 and the other on the southbound side of I-5. The treatment systems on both sides consists of eight 6,000 gallon septic tanks with leachfield disposal. The combined potable water use for the northbound and southbound SRRA is approximately 9,200 gpd, 90 percent of which is used for toilet flushing.

KARL E. LONGLEY ScD, P.E., CHAIR | PATRICK PULUPA, ESQ., EXECUTIVE OFFICER

The Discharger proposes to install an automated wastewater treatment system to provide disinfected tertiary-treated wastewater for non-potable use (toilet flushing). The Discharger expects to recycle approximately 80-90% of the wastewater flow with 10-20% (900-1,800 gpd) being dispersed to an existing leachfield on the northbound side of Interstate-5.

The proposed wastewater treatment system will consist of septic tanks, upflow anaerobic filters, backwashing filters, a textile media filter, a microfiber filter (Amiad AMF microfiber filter), ozone injection, and chlorine disinfection. Excess or off-spec wastewater can be diverted from the septic tanks, upflow filters, or after the chlorine contact basin to a leachfield (See Notice of Applicability Attachment B for a process flow diagram). The proposed system is designed to convey effluent from the upflow anaerobic filters to the leachfield when the flow exceeds the recycled water demand at the SRRA.

The disposal area consists of four leachfield areas, each with approximately 1,000 linear feet of dispersal laterals. The leachfield trenches consist of 2-foot wide trenches with perforated pipe set in a bed of sand at a minimum depth of 2 feet. The system is equipped with a dosing pump to alternate pumping between each of the dispersal areas. In a 17 November 2017 Title 22 Engineering Report, the Discharger states that the treatment system is designed to treat an average flow of 10,000 gallons per day with equalization capacity to accommodate peak flows up to 20,000 gallons per day and that the dispersal field (leachfield) has an area of 1 acre and is capable of receiving the full wastewater flow from the SRRA if needed.

RECYCLED WATER CONSIDERATIONS

Caltrans submitted a 17 November 2017 Title 22 Engineering Report for the recycled water project. The Title 22 Engineering Report was approved by a 12 February 2018 Division of Drinking Water (DDW) letter which provided the following recommended permit conditions, which includes Disinfection System Monitoring:

1. Caltrans shall comply with the approved Title 22 Engineering Report, dated 17 November 2017.
2. Any changes of what is described in the approved Title 22 Engineering Report, dated 17 November 2017, must be approved in writing by DDW.
3. The operation of the Amiad Filtration Systems AMF Wastewater Filter shall comply with the 8 June 2009 letter "Use of the Amiad Filtration Systems AMF Wastewater Filter to Comply With the California Water Recycling Criteria", including the following:
 - a. Each cassette shall be clearly marked with the TC-20 micron rating.
 - b. Filter loading rate shall not exceed 2.1 gpm/ft², and

- c. The turbidity of the influent to the AMF shall not exceed 5 NTU for more than 15 minutes and never exceed 10 NTU.
4. Minimum quarterly reporting requirements to DDW include the following information when the plant is operating:
 - a. total coliform running 7-day median calculation,
 - b. maximum daily total coliform reading for each month,
 - c. minimum daily chlorine residual when the plant is operating,
 - d. minimum daily chlorine contact time (CT),
 - e. average filter effluent turbidity (24 hour period), and
 - f. 95 percentile filter effluent turbidity (24 hour period)
 - g. daily maximum turbidity reading when the plant is operating
5. Within 24 hours, notification shall be given to the Central Valley Water Board, DDW, and the Merced County Environmental Health Department whenever the following occurs:
 - a. Effluent total coliform bacteria exceeds 240 MPN/100 ml,
 - b. Effluent turbidity exceeds 10 NTU,
 - c. A chlorine residual less than 2.5 mg/L is delivered into the distribution system, or
 - d. Discovery of any cross connections between the recycled water piping and the potable water piping.

POTENTIAL THREAT TO WATER QUALITY

The technical report states test pits in the area determined that groundwater is greater than 10 feet below ground surface and that no evidence of seasonal high groundwater was found during the excavations. The nearest well to the facility is located about 10,000 feet to the southeast of the SRRA, with depth to groundwater being 80 feet bgs. The nearest surface water is the California Aqueduct, which is located about 220 feet from the east border of the SRRA. The distance from the facility to both the nearest well and the California Aqueduct exceed the applicable minimum setback requirements described in *Table 3: Summary of Wastewater System Setbacks* of the General Order.

Soil data from the Natural Resources Conservation Service (NRCS), indicates that soils at the SRRA are identified as Los Banos-Pleito clay loams with 2 to 8 percent slopes.

The typical profile for the area is clay loam to a depth of 55 inches, with sandy/gravelly clay loam below 55 inches. The Facility's water source is the California Aqueduct.

The design flow for the treatment facility 10,000 gallons per day. Therefore, in accordance with the General Order no nitrogen limit is required.

MONITORING REQUIREMENTS

Monitoring requirements included in the following sections from Attachment C of the General Order are appropriate for this discharge:

- Septic Tank Monitoring;
- Subsurface Disposal Area;
- Disinfection System Monitoring; and
- Solids Disposal Monitoring.

CV-SALTS

The Central Valley Water Board adopted Basin Plan amendments incorporating new programs for addressing ongoing salt and nitrate accumulation in the Central Valley at its 31 May 2018 Board Meeting. These programs, once effective, could change how the Central Valley permits discharges of salt and nitrate.